
TRANSFORMER
Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Transformers,
Attention Mechanism
&
How GPT Learns Language
GPT
I
went
to
Mysore
yesterday
because
...
T

Transformers: The Architecture Behind GPT

Why Do We Need Transformers?

Before Transformers, AI used:

- **RNN** (Recurrent Neural Network)
- **LSTM** (Long Short-Term Memory)

Problems:

- Slow Training
- Difficulty remembering long sentences
- Cannot process words in parallel

Example:

I went to Mysore yesterday because ______

↓

To predict the next word, older models struggled to remember information far back in the sentence.

Transformers solve these problems by processing **all words together** and capturing long-range relationships effectively.

What is a Transformer?

A Transformer is a deep learning architecture introduced by Google in 2017.

Research Paper:
"Attention Is All You Need"

Transformers allow AI to:

- Understand relationships between words
- Process entire sentences simultaneously
- Learn long-range context

Transformer Architecture

Input Sentence
↓
Tokenization
↓
Embeddings
↓
Attention Mechanism
↓
Transformer Layers
↓
Prediction
↓
Output

What is Attention Mechanism?

Attention helps the model focus on important words in the input when making predictions.

Example:

Input Sentence:

| I | went | to | Mysore | yesterday | because | ... |

Model focuses more attention on important words like **"Mysore"**, **"yesterday"**

How GPT Learns Language

Pre-training on Massive Data
GPT is trained on billions of words from books, websites, articles, code, etc.

Next Token Prediction
GPT learns to predict the next most likely token based on the previous tokens.

I went to Mysore yesterday because it was a great ____ (next token)

Learning Patterns & Relationships
Through billions of predictions, GPT learns grammar, knowledge, facts, reasoning patterns, and more.

Fine-tuning & Alignment
Further training makes GPT more helpful, safe, and aligned with human preferences.

Generate Response
When you ask a question, GPT uses what it learned to generate the best possible response.

Why This Matters?

Transformers use Attention Mechanism to understand relationships between all words in a sentence in parallel. This enables GPT to remember long contexts, understand meaning better, and generate human-like responses.

Transformers are the foundation of modern LLMs like GPT, Claude, Gemini, Llama, and many more!

Transformers: The Architecture Behind GPT

Tokenization

Sentence:

I love learning AI.

Tokens:

I	love	learning	AI	.

Each token receives a unique ID.

Example:

Token		Token ID
I	→	101
love	→	102
learning	→	103
AI	→	104
.	→	105

Embeddings

- Computers do not understand words.
- Words are converted into numbers.

Example:

Mysore

becomes:

[0.52, 0.91, -0.33, 0.71]

This numerical representation is called an **Embedding.**

Word	Conversion (Embedding Layer)	Vector (Embedding)
Mysore →	→	[0.52, 0.91, -0.33, 0.71]

What is Attention?

Attention helps AI determine:

Which words are important?

Example

Sentence:

The animal didn't cross the road because **it** was **tired.**

Question:

Who was tired?

AI must understand:

it → **animal**

Attention helps identify this relationship.

Real-Life Example

Teacher asks:

Naveen teaches Python in Mysore.

Question:

Where does Naveen teach?

Attention focuses on:

Naveen → **Mysore**

Key Takeaway: Tokenization breaks text into tokens, Embeddings convert tokens into numbers, and Attention helps the model focus on what matters most to understand context.

These three concepts are the foundation of how Transformers and GPT understand and generate human language.

Transformers: The Architecture Behind GPT

Self-Attention

★ The most important concept in Transformers.

Each word looks at **every other word.**

Example:

I love AI

Word: **AI** looks at:

I | love | AI

and decides which words are most relevant.

Visualization

I

love

AI

💡 Every word interacts with every other word.

Query, Key, Value (QKV)

Attention uses three vectors:

Q Query (Q)	What am I looking for?
K Key (K)	What information do I have?
V Value (V)	Actual information to return.

Example

Student asks: Who is the Principal?

Query: Principal

School Database:

Keys = Names	Values = Information
Principal	In charge of school
Teacher	Teaches subjects
Student	Learns subjects
...	...

→ Matching **Key** returns **Value.**

✓ This is the same principle used inside GPT.

Multi-Head Attention

★ Instead of looking at one relationship: Transformer looks at **many relationships simultaneously.**

Example:

Naveen teaches Python in Mysore.

Head 1
Focus: Who is doing the action?
Naveen ⟷ teaches
(Relationship captured)

Head 2
Focus: What is being taught?
Python ⟷ teaches
(Relationship captured)

Head 3
Focus: Where is the action happening?
Mysore ⟷ teaches
(Relationship captured)

This improves understanding by capturing multiple relationships at the same time.

Key Takeaway

Self-Attention
Each word looks at every other word to understand relationships.

QKV
Query, Key, Value work together to find the most relevant information.

Multi-Head Attention
Transformer looks at many relationships in parallel for a deeper and richer understanding.

These mechanisms allow Transformers (like GPT) to understand context powerfully and generate human-like responses.

Transformers: The Architecture Behind GPT

Transformer Layers

A Transformer contains many layers.

Example:

Input
↓
Layer 1
↓
Layer 2
↓
Layer 3
↓
...
↓
Output

Each layer learns more complex patterns.

Deeper layers capture higher-level and more abstract patterns.

How GPT Learns Language

GPT learns by predicting the next word.

Example

Training Data: The capital of India is → **Target:** New Delhi

GPT predicts: Mumbai — Wrong Prediction ✗

Model adjusts weights.

Again: The capital of India is → **Predicts:** New Delhi

Correct Prediction ✓

This process repeats billions of times.

Training Process

Internet Data (Massive Text from Web)
↓
Tokenization (Convert text to tokens)
↓
Transformer Network (Deep Learning Model)
↓
Prediction (Predict next word)
↓
Error Calculation (Compare with correct word)
↓
Weight Update (Improve the model)
↓
Repeat Billions of Times (Until the model is highly accurate)

Why GPT is Powerful

Because it has learned patterns from:

- Books
- Articles
- Websites
- Research Papers
- Documentation
- Conversations

It doesn't memorize everything.
It learns patterns.

Key Takeaways

- Transformers use multiple layers to learn complex representations.
- GPT learns by predicting the next word and improving through repetition.
- Training involves tokenization, prediction, error calculation, and weight updates.
- GPT is powerful because it learns patterns from massive amounts of diverse data.

Transformers: The Architecture Behind GPT

Key Interview Questions

Q1 What is a Transformer?

A A deep learning architecture that uses attention mechanisms to process language efficiently.

Q2 What is Self-Attention?

I love AI

A A mechanism allowing each word to focus on other relevant words in the sentence.

Q3 What are Query, Key, and Value?

Query (Q)	Key (K)	Value (V)
What am I looking for?	What information do I have?	Actual information to return.

A Vectors used to calculate attention scores and determine relevant information.

Q4 Why are Transformers better than RNNs?

Faster training | Parallel processing | Better long-context understanding

A

- Faster training
- Parallel processing
- Better long-context understanding

Q5 How does GPT learn?

Predict next token → Compare with actual token → Calculate error → Update parameters → Repeat billions of times

A By predicting the next token repeatedly during training and adjusting its parameters based on prediction errors.

Pro Tip

Understanding these core concepts is key to mastering how GPT and other LLMs work. Practice with examples and visualizations to strengthen your understanding!